

TOPS TECH Task 9 (CS)

Task-9 Vulnerability Scanning Submitted

- 1. Install and run the free version of the OpenVAS or Nessus vulnerability scanner on your system, and perform a basic scan of your own computer or a local virtual machine.**

Ans:

Vulnerability Scanning Using Nessus Essentials

Objective:

- To identify security vulnerabilities, misconfigurations, and potential risks present on the local system using Nessus Essentials Vulnerability Scanner.
- Open Nessus through the web interface using:
<https://localhost:8834>
- Create a new scan by selecting "Basic Network Scan".

Enter the scan details:

- Name: Localhost Scan ○ Target: 127.0.0.1 ○ Save the scan configuration and launch the scan.
- Wait for the scan to complete and review the generated report.

Observations:

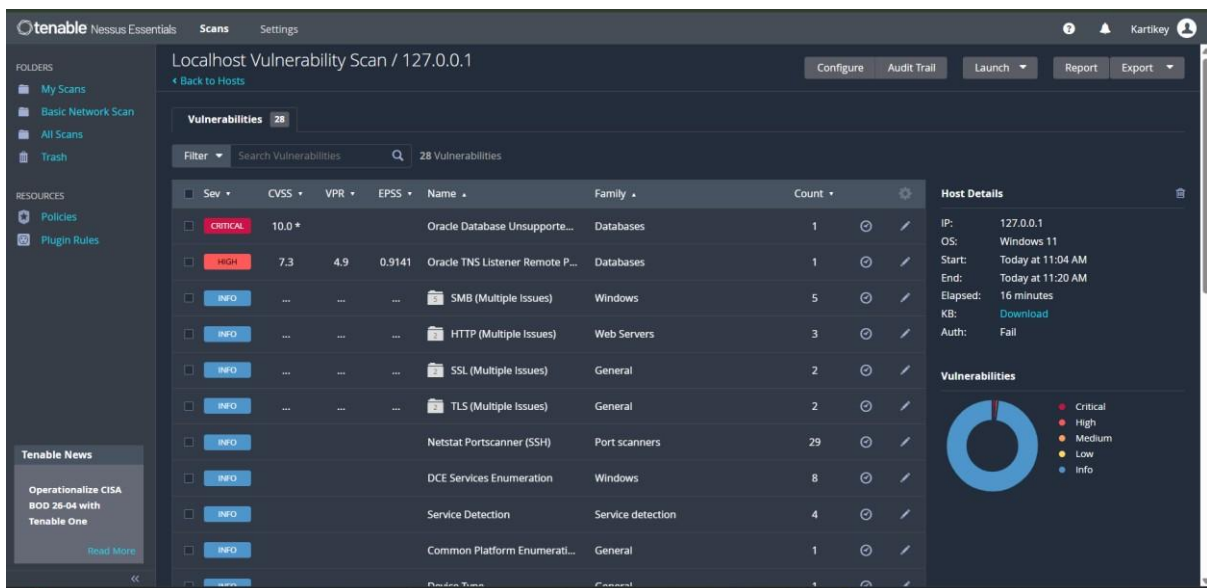
- Nessus successfully scanned the local system and identified active services and system configurations.
- The scanner checked installed software, operating system configuration, open ports, and security settings.
- Several informational items and warnings were detected during the scan.
- Nessus generated a detailed report showing vulnerabilities categorized by severity levels such as Critical, High, Medium, Low, and Informational.

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- The scan report also provided recommendations for improving system security.

Sample Vulnerabilities Identified:

- Severity Description
 - Medium Outdated software version detected
 - Low SSL/TLS configuration warning
 - Info Open network service identified
- ⇒ Nessus Essentials was successfully installed and configured.
- ⇒ A vulnerability assessment of the local system was performed successfully.
- ⇒ The scan report identified system information, open services, and potential security risks that can be used to improve the overall security posture of the system.



2. Scan your Wi-Fi router's IP address using Nmap and list all open ports detected.
Hint: Use the

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command 'nmap -v [router_ip]' in your terminal or command prompt.

Ans:

Wi-Fi Router Port Scanning Using Nmap

Objective:

- To identify open ports and associated network services running on the Wi-Fi router using the Nmap network scanning tool.

Command:

→ ipconfig

- The ipconfig command is used to identify the Default Gateway address, which represents the IP address of the Wi-Fi router.

Example:

- Default Gateway : 192.168.31.1

Command:

- nmap -sV 192.168.31.1

- The nmap -sV command scans the target device and attempts to determine the services and versions running on open ports.

Observations: ○ Nmap successfully scanned the Wi-Fi router and identified several open ports.

- The detected ports were associated with router management and network communication services.
- Open ports indicate services currently accepting network connections.

The following open ports were identified:

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<u>Port Number</u>	<u>Protocol</u>	<u>State</u>	<u>Service</u>
53	TCP	OPEN	domain
80	TCP	OPEN	http
443	TCP	OPEN	https
8080	TCP	OPEN	http-proxy
8443	TCP	OPEN	https-alt

Port Description:

<u>Port Number</u>	<u>Description</u>
53	Domain Name Service (DNS)
80	Web-based Router Management Interface
443	Secure Router Management Interface
8080	Alternative HTTP Management Service
8443	Alternative Secure HTTPS Management Service

- The router was found to be running network management services required for administration and internet connectivity.
 - Ports 8080 and 8443 indicate additional web-based management interfaces available on the router.
 - No critical issues were observed during the basic port scan.
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- ⇒ The Wi-Fi router was successfully scanned using Nmap.
 - ⇒ Open ports and associated services were identified and documented.

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- ⇒ The scan provided useful information regarding the network services exposed by the router and helped understand the network attack surface.

3. Analyze the scan report from your vulnerability scanner and identify at least 3 vulnerabilities or warnings it reports; for each, write a oneline description of what the risk is.

Ans:

Vulnerability Analysis Using Nessus Scan Report

Objective: ○ To analyze the Nessus vulnerability scan report and identify vulnerabilities and security warnings present on the local system.

Tool Used:

- Nessus Essentials Vulnerability Scanner

Target:

- 127.0.0.1 (Localhost)

Observations:

- Nessus successfully scanned the local system and identified 28 vulnerabilities and informational findings.
- The scan report categorized findings into Critical, High, and Informational severity levels.
- Several issues related to Oracle services, SMB services, HTTP services, SSL/TLS configurations, and service enumeration were detected.

The following vulnerabilities/warnings were identified:

1. Oracle Database Unsupported Version

- Severity: Critical ○ Risk:
- ⑨ The Oracle Database version installed on the system is no longer supported by the vendor.

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- ⑨ Unsupported software does not receive security updates and may contain known vulnerabilities that attackers can exploit.

Recommendation:

- Upgrade to a supported Oracle Database version and apply the latest security patches.

2. Oracle TNS Listener Remote Information Disclosure

- Severity: High ○ Risk:
 - ⑨ The Oracle TNS Listener may disclose sensitive information about the database environment, which can assist attackers in gathering intelligence for further attacks.

Recommendation:

- Restrict listener access and apply Oracle security updates.

3. SMB (Multiple Issues)

- Severity: Informational ○ Risk:
 - ⑨ SMB services may expose information about the system configuration and network resources, increasing the attack surface if improperly

Recommendation:

- Disable unnecessary SMB services and restrict access using firewall rules.

4. HTTP (Multiple Issues)

- Severity: Informational ○ Risk:
 - ⑨ Web services may reveal server information, configuration details, or application fingerprints that can aid attackers during reconnaissance.

Recommendation:

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- Minimize information disclosure and keep web services updated.

4. SSL/TLS (Multiple Issues)

- Severity: Informational

Risk:

- ⑨ Weak SSL/TLS configurations may expose encrypted communications to interception or downgrade attacks.

Recommendation:

- Use strong encryption protocols and disable outdated SSL/TLS versions.

Summary Table:

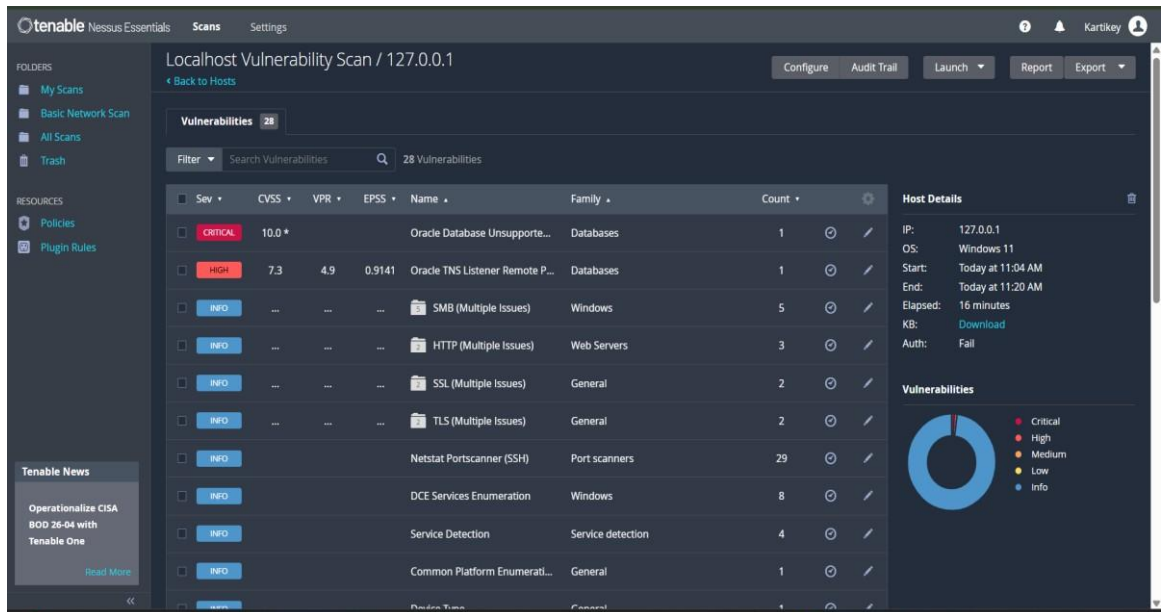
Vulnerability Severity Risk Description ○ Oracle Database

Unsupported Version Critical Unsupported software may contain exploitable vulnerabilities.

- Oracle TNS Listener Information Disclosure High Sensitive system information may be exposed.
- SMB Multiple Issues Informational May reveal system and network details.
- HTTP Multiple Issues Informational May expose web server information.
- SSL/TLS Multiple Issues Informational Weak encryption settings may affect communication security.

- ⇒ The Nessus scan report was successfully analyzed.
- ⇒ One Critical vulnerability, one High severity vulnerability, and multiple informational security warnings were identified.
- ⇒ Appropriate recommendations were suggested to reduce security risks and improve the overall security posture of the system.

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4. Pick any app you use daily (like Instagram, Zomato, or Paytm) and research one real vulnerability that was found in it in the past; write a short summary of what the vulnerability was and how it could have been detected by scanning tools.

Ans:

Research on a Real Vulnerability in WhatsApp

Objective:

- To study a real-world security vulnerability discovered in a widely used application and understand its impact and detection methods.

Application Selected:

- WhatsApp

Vulnerability Name:

- Pegasus Spyware Vulnerability (2019)

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- In 2019, a serious vulnerability was discovered in WhatsApp that allowed attackers to install Pegasus spyware on a victim's mobile device.
- The vulnerability existed in WhatsApp's voice calling feature.
- An attacker could exploit this flaw by placing a specially crafted WhatsApp call to the target device.
- In some cases, the spyware could be installed even if the victim did not answer the call.

Impact:

- Attackers could gain unauthorized access to sensitive information stored on the device.
- Personal messages, emails, contacts, location data, and other private information could potentially be monitored.
- The vulnerability affected both Android and iOS devices.

Risk Analysis:

Risk Type Description:

- Privacy Risk User data could be accessed without permission.
- Information Disclosure Sensitive information could be exposed.
- Remote Exploitation Attackers could compromise a device remotely.
- Surveillance Risk Spyware could monitor user activities secretly.

How Scanning Tools Could Detect It: ○ Vulnerability scanners can identify outdated application versions that contain known security flaws.

- Security assessment tools can compare installed software versions against vulnerability databases.
- Mobile security scanners can detect suspicious application behaviour and unusual network activity.
- Regular security audits and vulnerability assessments help identify software that requires security updates.

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Control:

- Update WhatsApp to the latest version.
 - Install security patches released by the vendor. ○ Avoid using outdated software versions.
 - Enable device security features and keep the operating system updated.
- ⇒ A real-world vulnerability affecting WhatsApp was successfully studied.
- ⇒ The Pegasus spyware vulnerability demonstrated how a software flaw can lead to serious privacy and security risks.
- ⇒ Regular updates, vulnerability scanning, and security monitoring can help prevent such attacks and improve overall cybersecurity.

**5. Use ChatGPT or Copilot to generate a basic bash or PowerShell script that runs Nmap on a list of 3 IP addresses and saves the results to separate files; copy and run the script, and share the output files.

Hint: Ask the AI tool for a script that loops through IPs and uses Nmap for each.**

Ans:

Automated Nmap Scanning Using PowerShell Script

Objective:

- To automate Nmap scanning of multiple IP addresses using a PowerShell script and store the results in separate output files.

Tool Used:

- Nmap
- Windows PowerShell

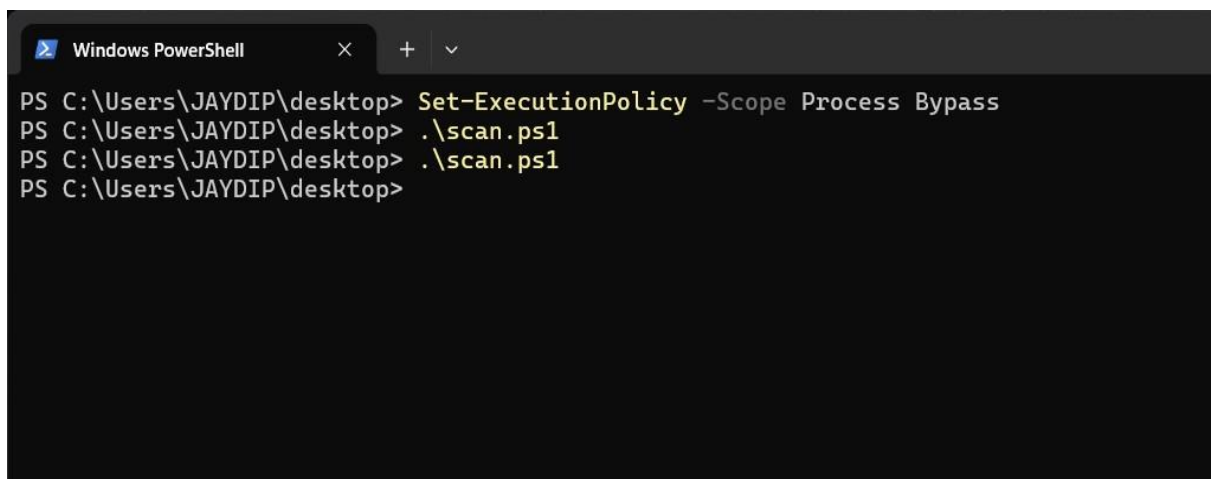
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PowerShell Script:

```
$ips = @(
"192.168.31.1",
"127.0.0.1",
"192.168.31.165"
)
foreach ($ip in $ips)
{
nmap -F $ip > "$ip.txt"
}
```

Script Explanation:

- The script stores three IP addresses in an array.
- The foreach loop processes each IP address one by one.
- Nmap performs a fast scan (-F) on each target.
- The output of each scan is redirected to a separate text file.
- The generated files are named according to the scanned IP address.



```
Windows PowerShell
PS C:\Users\JAYDIP\desktop> Set-ExecutionPolicy -Scope Process Bypass
PS C:\Users\JAYDIP\desktop> .\scan.ps1
PS C:\Users\JAYDIP\desktop> .\scan.ps1
PS C:\Users\JAYDIP\desktop>
```

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Observations:

- The script successfully automated the scanning process.
 - Each target was scanned individually. ○ Scan results were stored in separate files for easier analysis.
 - The generated reports contained information about open ports and available network services.
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- ⇒ A PowerShell script was successfully generated using ChatGPT.
 - ⇒ Nmap scans were executed automatically on multiple IP addresses.
 - ⇒ Individual scan reports were generated and stored in separate output files, demonstrating basic automation of network scanning tasks.